

Mass Loss Study Report

Foam Specimens — Evaporative Mass Loss under Solar Simulation

March 18–19, 2026

Experimental Procedure

Specimen Preparation: Pre-made foam specimens were cut to $3\text{ cm} \times 3\text{ cm}$ and placed back into the oven prior to testing.

Mass Measurement:

1. A piece of measuring paper was placed on a tared scale and re-tared.
2. A cut foam specimen was placed in the centre of the paper and its mass was recorded.
3. The specimen was removed and the scale was re-zeroed.
4. A PLA evaporation disk was placed on the balance (not tared).
5. Using a dropper, 1 mL of deionised (DI) water was added to the evaporation disk and the combined weight was noted in the lab notebook.

Specimen Rinsing & Setup: The weighed foam specimen was rinsed thoroughly with deionised water until all loose particles were removed. It was then placed into the evaporation disk and the assembly was positioned on the solar simulator.

Data Acquisition:

1. Sensors (SHT4x and MLX90614) were connected and logged via the Excel sheet.
2. PuTTY was opened on **COM3**, **COM4**, and **COM5/COM6** to capture serial output.
3. A light sensor was used to obtain an estimated irradiance temperature.
4. After 3600 s, the terminal output was copied to clipboard and entered into the Excel spreadsheet.

Post-Test Measurement & Longevity Study: Specimens were *not* discarded after testing. Each foam was re-weighed and the new mass recorded in the lab notebook. Specimens are set aside for repeated testing to evaluate foam reusability and longevity.

Mass Loss Rate Formula:

$$\dot{m} [\text{kg/m}^2/\text{hr}] = \frac{\text{Slope [g/s]}}{1000} \times \frac{(100 \text{ cm/m})^2}{9 \text{ cm}^2} \times 3600 \text{ s/hr}$$

Global Observations:

- All samples with complete data returned $R^2 > 0.991$, confirming strong linear evaporation behaviour throughout the 3600 s window.

- A warm-up transient is visible in most samples during the first 5–10 minutes where evaporation starts slower then accelerates — this is a simulator heating effect, not a material property.
- No discontinuities or non-monotonic weight increases were detected in any sample across all batches. Data quality is clean throughout.

Batch 1 — 5% Cement Mix (March 18)

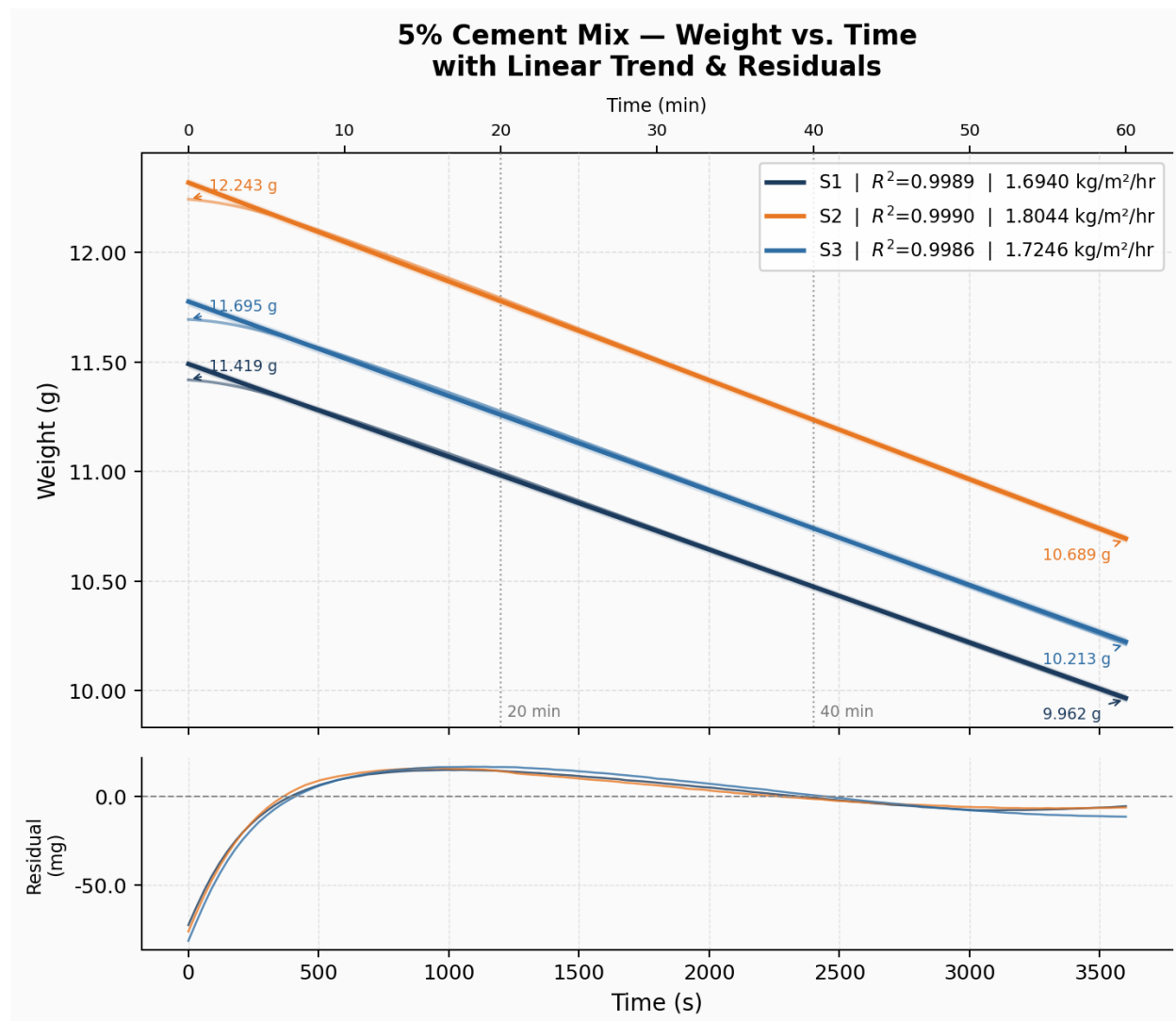


Figure 1: *
Weight vs. time — 5% cement samples (S1, S2, S3)

Sample	Initial (g)	Final (g)	Δm (g)	Loss (kg/m ² /hr)
S1	11.4194	9.9619	1.4575	1.6940
S2	12.2434	10.6894	1.5540	1.8044
S3	11.6953	10.2132	1.4821	1.7246
Average	—	—	1.4979	1.7410

Observations:

- R^2 : S1 = 0.9989, S2 = 0.9990, S3 = 0.9986 — all very strong.
- Initial weights vary by 0.82 g (11.42–12.24 g). Despite this, evaporation rates are tightly

grouped (slopes differ $<7\%$).

- All curves strictly monotonically decreasing with no artefacts.

Batch 2 — 20% Cement Mix (March 18)

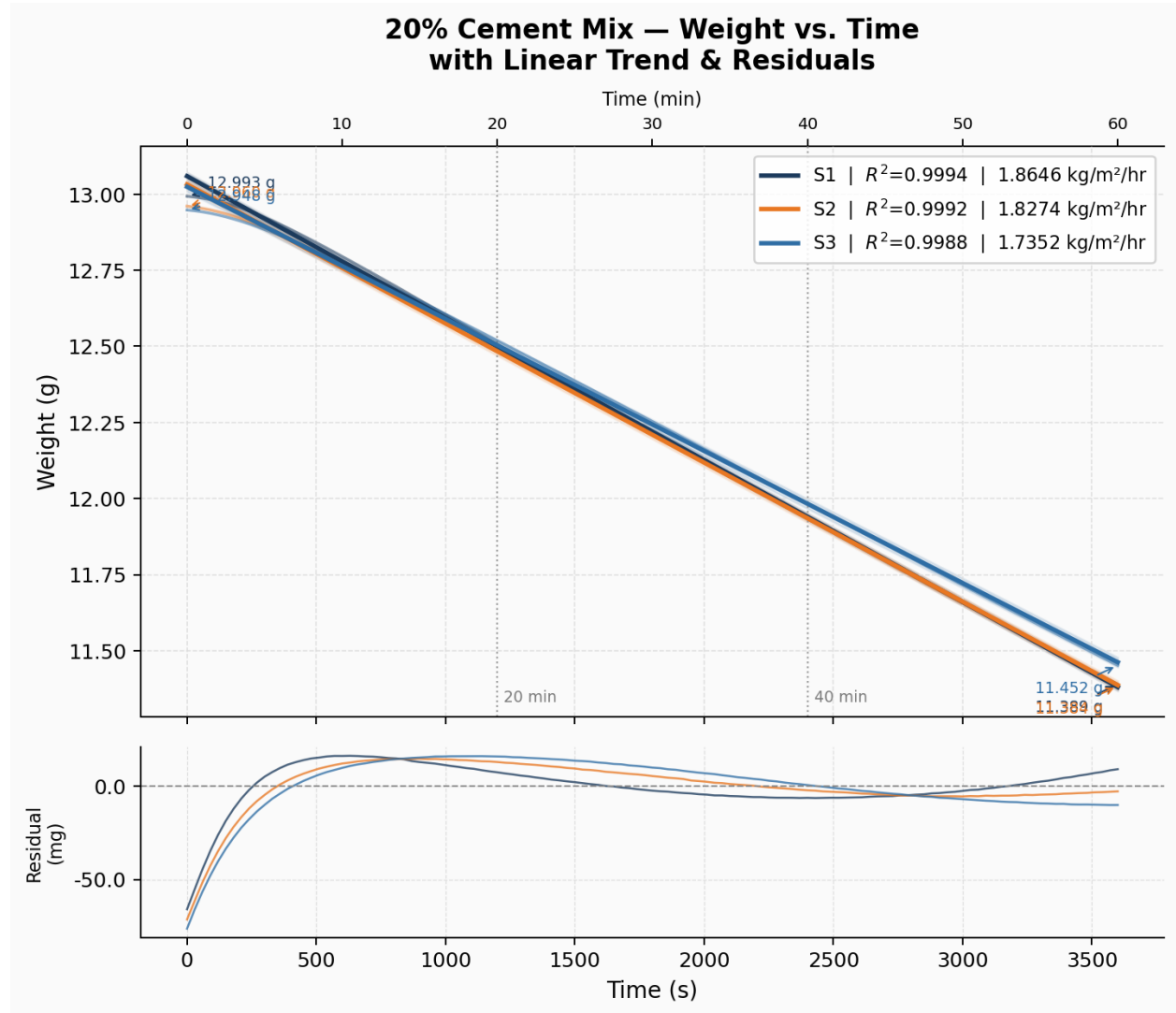


Figure 2: *
Weight vs. time — 20% cement samples (S1, S2, S3)

Sample	Initial (g)	Final (g)	Δm (g)	Loss ($\text{kg}/\text{m}^2/\text{hr}$)
S1	12.9928	11.3894	1.6034	1.8646
S2	12.9603	11.3840	1.5763	1.8274
S3	12.9476	11.4519	1.4957	1.7352
Average	—	—	1.5585	1.8091

Observations:

- R^2 : S1 = 0.9994, S2 = 0.9992, S3 = 0.9988 — slightly better linearity than Batch 1.
- **Exceptional specimen consistency**: all three initial weights within 0.045 g of each

other (12.95–12.99 g).

- Highest average mass loss rate of all four batches ($1.8091 \text{ kg/m}^2/\text{hr}$).

Batch 3 — 90% Graphite / 10% Cement (March 18)

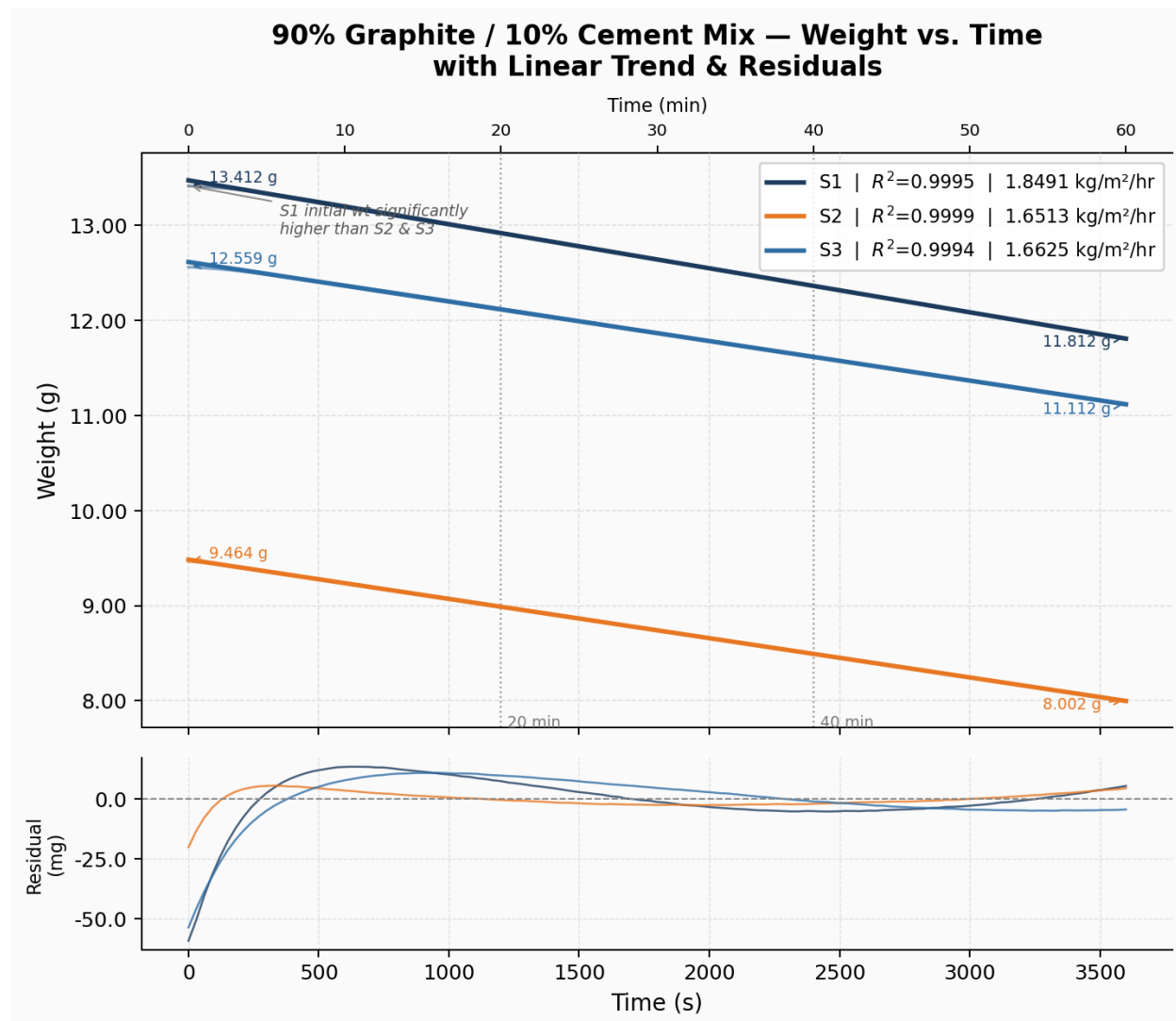


Figure 3: *
Weight vs. time — 90% graphite / 10% cement samples (S1, S2, S3)

Sample	Initial (g)	Final (g)	Δm (g)	Loss (kg/m ² /hr)
S1	13.4118	11.8124	1.5994	1.8491
S2	9.4636	8.0021	1.4615	1.6513
S3	12.5591	11.1122	1.4469	1.6625
Average	—	—	1.5026	1.7210

Observations:

- R^2 : S1 = 0.9995, S2 = 0.9999, S3 = 0.9994 — **highest linearity of all batches**. S2 is remarkably consistent.

- **Major flag: initial weights vary by 3.95 g** (9.46–13.41 g). Despite this, Δm values converge (1.45–1.60 g), suggesting rate is governed by surface area rather than total mass.
- No discontinuities or weight increases detected.

Batch 4 — 5 wt% Cement + Graphite (March 19)

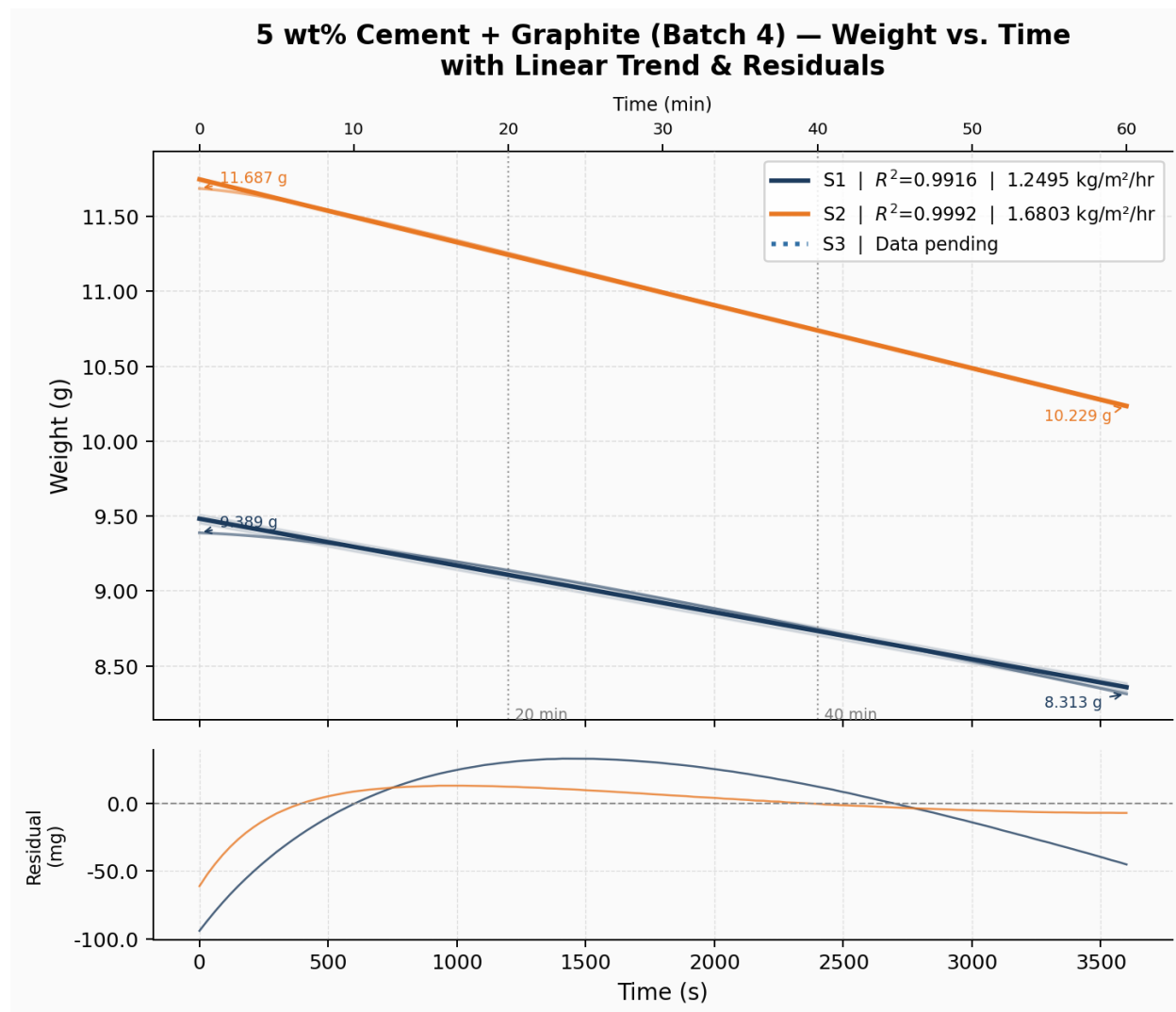


Figure 4: *

Weight vs. time — 5 wt% cement + graphite samples (S1, S2, S3)

Sample	Initial (g)	Final (g)	Δm (g)	Loss (kg/m ² /hr)
S1	9.3888	8.3132	1.0756	1.2495
S2	11.6870	10.2288	1.4582	1.6803
S3	9.3754	7.9599	1.4155	1.6174
Average	—	—	1.3164	1.5157

Observations:

- R^2 : S1 = 0.9916, S2 = 0.9992, S3 = 0.9998. S3 is excellent; S1 remains the lowest R^2 across all batches.

- **S1 anomalous warm-up transient confirmed:** slope accelerates -70% from first to last third of the run. S3 by contrast shows only -3.1% acceleration — the most thermally stable single sample in the entire dataset.
- S2 (1.6803) and S3 (1.6174) are close and consistent. With S1 excluded as an outlier, the effective rate would be $\sim 1.649 \text{ kg/m}^2/\text{hr}$ — much more in line with other batches.
- Ambient conditions on March 19 were cooler (21.7°C vs. $\sim 30^\circ\text{C}$) and more humid (61%) than March 18, which likely suppressed evaporation rates across the whole batch.

Reliability Study — 20% Cement Re-test (Same Specimens)

Reliability Test Procedure:

Per the instructions of Dr. Boymelgreen, the following protocol was followed for the reliability run. Note: insufficient samples were available to compare dry re-wetted vs. continuously wetted foam, so all specimens were kept continuously wetted between runs.

1. Each foam specimen was placed individually in a petri dish with deionised (DI) water — enough to maintain moisture without waterlogging — and soaked for a minimum of 1 hour to ensure full wetting prior to testing.
2. Using a fresh piece of weighing paper on a tared scale, the wet foam specimen was weighed and the mass recorded in the lab notebook.
3. The scale was cleared and re-tared. The PLA evaporation disk was placed on the balance and 1 mL of DI water was added via dropper. The combined weight was recorded.
4. The wetted foam specimen was placed into the evaporation disk containing the DI water, and the total combined weight was recorded.
5. The assembly was transferred to the solar simulator and the 3600s data acquisition was started as per the standard procedure.

Sample	Initial (g)	Final (g)	Δm (g)	Loss (kg/m ² /hr)
Original S1	12.9928	11.3894	1.6034	1.8646
Original S2	12.9603	11.3840	1.5763	1.8274
Original S3	12.9476	11.4519	1.4957	1.7352
Original Avg	12.9669	—	1.5585	1.8091
Reliability S1	12.8536	11.5895	1.2641	1.4670
Reliability S2	12.4964	11.1064	1.3900	1.6101
Reliability S3	12.4886	11.1014	1.3872	1.5953
Reliability Avg	12.6129	—	1.3471	1.5575
Change vs. Original	−0.354 g	—	−0.211 g	−13.9%

Observations:

- All three reliability samples return strong R^2 values (S1: 0.9987, S2: 0.9992, S3: 0.9992), confirming the foam maintains clean linear evaporation behaviour after one full use cycle.
- **The average mass loss rate dropped 13.9%** (1.8091 $\rightarrow$ 1.5575 kg/m²/hr) on re-test. With three samples now confirming this trend, it is no longer attributable to a single outlier.
- **S2 and S3 are very consistent with each other** (1.6101 vs. 1.5953 kg/m²/hr, a difference of only 0.9%), suggesting reliable and repeatable behaviour for those specimens. S1 sits lower at 1.4670, which may reflect environmental conditions at the time it was tested or slight specimen-level variation.

- **Initial weights declined across all three** (avg 12.97 g $\rightarrow$ 12.61 g, a drop of 0.354 g), indicating the foam holds less water after one use cycle even when kept wet between runs.
- **Δm declined** from 1.559 g to 1.347 g on average — less total water evaporated per hour, consistent with reduced surface water availability.
- **The rolling rate overlay** confirms the reliability samples track consistently below the originals throughout the full run, ruling out any start-up or end-of-run artefact as the cause.
- **Interpretation:** a 13.9% reduction after one cycle is meaningful. Some portion is likely environmental (ambient conditions varied between test days), but the consistent decline in initial weight across all three specimens points to genuine foam behaviour change — likely a reduction in effective pore volume available for water storage and transport. Further cycles will clarify whether this decline continues, stabilises, or accelerates.

Comparison — All Four Batches

Metric	B1: 5% Cem	B2: 20% Cem	B3: 90%Gr	B4: 5wt%+Gr
Avg. Δm (g)	1.498	1.559	1.503	1.316
Avg. loss (kg/m ² /hr)	1.7410	1.8091	1.7210	1.5157
Avg. R^2	0.9988	0.9991	0.9996	0.9969
Initial wt. spread (g)	0.824	0.045	3.948	2.311
Date	Mar 18	Mar 18	Mar 18	Mar 19

Observations:

- **Ranking (all batches):** B2 (1.8091) > B1 (1.7410) > B3 (1.7210) > B4 (1.5157). The gap between B1–B3 is small (<5%); B4 sits noticeably lower, largely attributable to different ambient conditions on March 19 and S1’s anomalous warm-up.
- If S1 of Batch 4 is excluded as an outlier, B4’s effective average rises to ~ 1.649 kg/m²/hr — within 4% of Batch 3.
- **Batch 3 has the best average R^2** (0.9996), suggesting graphite content promotes the most thermally uniform evaporation surface.
- The rolling rate panel confirms the simulator warm-up effect is present in all batches. Batch 4 S1 stands out clearly with its slow start.
- The reliability study (13.9% rate reduction after one cycle) raises an important question for the longevity study: how much of this is environmental and how much is foam degradation? Further cycles will clarify.

Environmental Conditions

Parameter	B1: 5% Cem	B2: 20% Cem	B3: 90%Gr	B4: 5wt%+Gr
Ambient temp (°C)	30.2	30.5	30.4	21.7
Rel. humidity (%)	54.7	53.8	47.2	61.1
Surface temp (°C)	60.0	59.3	59.8	33.6

Note: Batch 4 and the Reliability run were conducted on March 19 under noticeably cooler and more humid conditions. Direct comparisons to March 18 batches should account for this environmental difference.

Next Steps

- Re-run reliability test under matched ambient conditions to isolate foam degradation from environmental effects
- Continue repeat cycles on all specimens for longevity study
- Investigate S1 Batch 4 anomalous warm-up — consider pre-warming simulator before placing specimen
- Investigate large initial weight spread in Batch 3 graphite specimens
- Extend test duration beyond 1 hour to capture any non-linear behaviour
- Normalise ambient conditions across batches for fair comparison

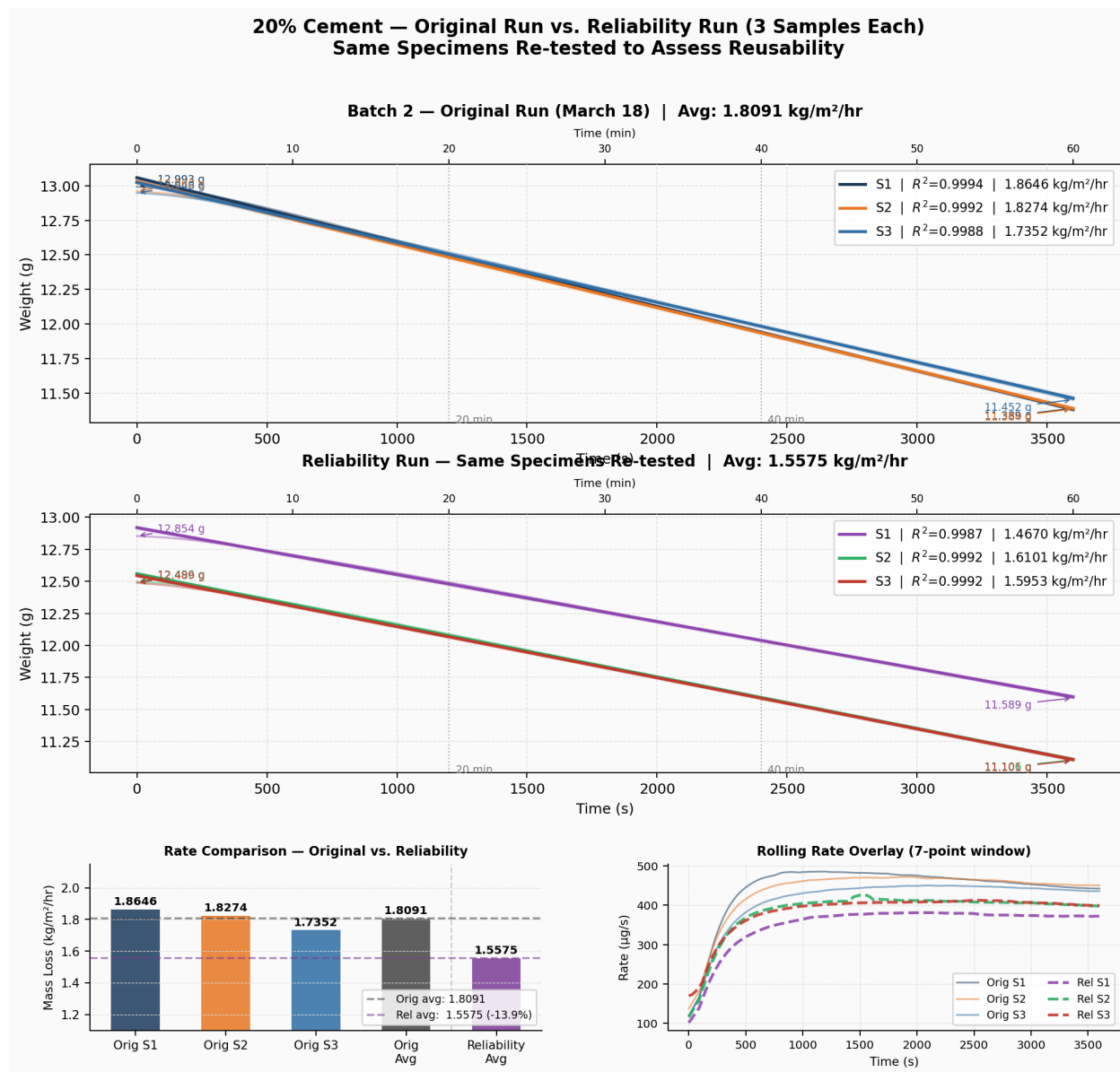


Figure 5: *

Top: original batch (S1, S2, S3). Middle: reliability batch (S1, S2, S3) re-tested on the same specimens. Bottom left: grouped bar chart comparing each sample pair side by side. Bottom right: rolling rate overlay.

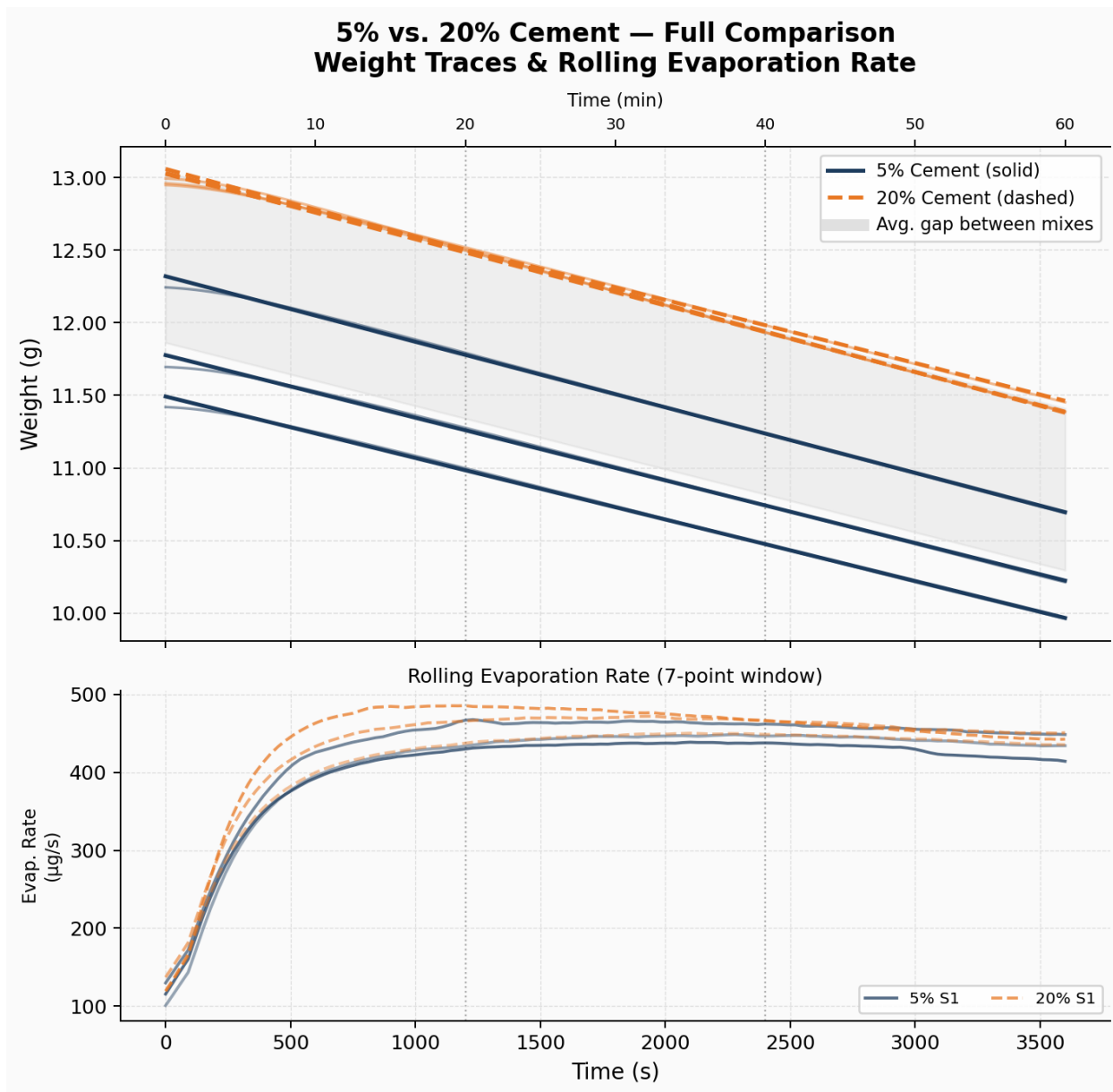


Figure 6: *
All batches overlaid with rolling evaporation rate panel